

Ainesh Sabharwal

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science, College of Computing, Data Science, and Society.

First-year student; classified as sophomore due to AP and transfer credits.

Relevant Coursework: Linear Algebra and Differential Equations, Computer Science 61A, Data Structures and Algorithms (Java)

Bridgeland High School

Class Rank 2 of 921. GPA: 4.0/4.0 unweighted.

Berkeley, CA

Expected 2029

Cypress, TX

2026

SKILLS

Languages: Java, TypeScript, Python, Swift

Web and Full-Stack: React, HTML/CSS, full-stack web applications, dashboards and analytics

AI / ML: Retrieval-Augmented Generation (RAG), on-device LLM inference, document summarization

CS Foundations: stacks, queues, ring buffers, priority queues, trees, recursion, greedy algorithms

Tools: Git, GitHub, Eclipse

EXPERIENCE

Undergraduate Research Apprentice

Sep 2026 - Present

Local Laws Observatory | UC Berkeley URAP

- Contributing to research infrastructure for collecting, organizing, searching, and analyzing U.S. local laws.
- Project work includes legal document data quality, search and retrieval, and AI evaluation against source text.

Lead Instructor

Oct 2025 - Mar 2026

iCode

- Led coding and STEM classes for K-12 students, teaching programming fundamentals through hands-on projects.
- Collaborated with the lead director to rebuild the standardized curriculum after identifying that the original material was above students' level.

Student Researcher

Aug 2024 - May 2025

Algoverse AI Research Program

- Conducted applied research on Retrieval-Augmented Generation (RAG) systems for grounded question answering.
- Explored retrieval, embeddings, and prompting strategies to improve answer accuracy against source documents.
- Developed AI-generated question pipelines and evaluated output faithfulness until the project reached a stage requiring faculty involvement to assess generated questions.

PROJECTS

Data Structures and Algorithms Labs github.com/sabharwalainesh

Java

- Implemented Huffman compression, a 20 Questions binary decision tree with file-backed learning, real-time audio synthesis via Karplus-Strong on a custom ring buffer, and a Java Swing image editor with undo/redo backed by a custom array-based stack.

HuddleHub

Full-Stack Local Business Platform, TypeScript, FBLA 2025-26

- Built a full-stack web application to browse, search, review, and bookmark local businesses, including an owner dashboard to manage listings and track engagement analytics; placed 2nd in Texas at the FBLA state competition.

Bridgeland Arena Hub

Event Ticketing and Box Office Platform, React/TypeScript, FBLA 2024-25

- Built a ticketing platform for an event venue covering event browsing, seat selection, purchase flow, and a staff-side dashboard for managing events and tracking sales; placed 3rd in the nation at the FBLA National Leadership Conference.

ApathyAI github.com/qurashisohaib/ApathyAI

On-Device AI Copilot, Python, Contributor

- Contributed to an open-source desktop AI assistant that runs fully on-device; implemented on-screen context reading to auto-summarize documents and suggest spreadsheet and form fills.

AWARDS & HONORS

- 3rd Place, Website Design, FBLA National Leadership Conference (2025)
- 2nd Place, Texas State, FBLA Website Design Competition (2025-26)
- Eagle Scout, Boy Scouts of America
- Black Belt, Taekwondo